# Approximation of Fixed Point Mappings

Pathak (2018) – Chapter 6e
Applications of Fixed Point Theorems

Lecture Notes

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# Contents

# Fixed Point Mappings and Iteration I

## Definition (Fixed Point)

Let $(X, \|\cdot\|)$ be a Banach space and $T : D \subseteq X \to X$ be a mapping. A point $x^* \in D$ is called a **fixed point** of $T$ if

$$T(x^*) = x^*.$$

## Definition (Fixed Point Approximation)

Given a mapping $T$ with a fixed point $x^*$, an **approximation sequence** $\{x_n\}$ is defined recursively by

$$x_{n+1} = T(x_n), \quad n = 0, 1, 2, \ldots$$

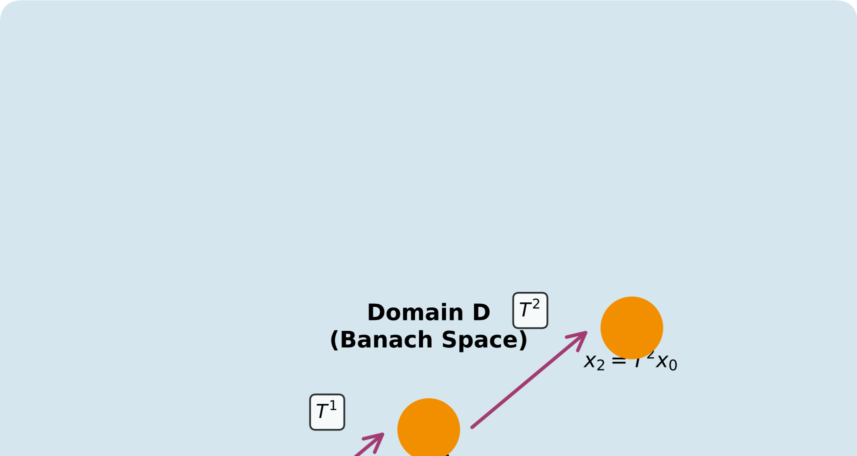
where $x_0 \in D$ is an initial guess.

**Theorem (Existence and Uniqueness)**

*If $T : D \to D$ is a continuous mapping on a compact set $D$ in a Banach space, then $T$ has at least one fixed point.*

# Fixed Point Mappings and Iteration III

**Fixed Point Mapping: Sequence Generation**



# Basic Iterative Scheme

## Picard Iteration

The most basic approach to approximate a fixed point is the Picard iteration:

$$x_{n+1} = T(x_n), \quad n = 0, 1, 2, \ldots$$

**Key Questions:**

1. Does the sequence $\{x_n\}$ converge?
2. If it converges, does it converge to a fixed point?
3. What is the rate of convergence?
4. How many iterations are needed for a desired accuracy?

The convergence depends on properties of $T$ (continuity, contractiveness, etc.) and the structure of the domain $D$.

# Contraction Mappings

## Definition (Contraction Mapping)

A mapping $T : D \subseteq X \to X$ is a **contraction mapping** (or $\lambda$-contraction) if there exists a constant $0 \leq \lambda < 1$ such that

$$\|T(x) - T(y)\| \leq \lambda\|x - y\| \quad \text{for all } x, y \in D.$$

The constant $\lambda$ is called the **contraction rate**.

## Properties

Every contraction is continuous.

A contraction is uniformly continuous.

The contraction rate measures how fast distances shrink.

# Banach Contraction Mapping Theorem

## Theorem (Banach (1922))

*Let $(X, \|\cdot\|)$ be a complete metric space and $T : X \to X$ be a contraction mapping with contraction rate $0 \leq \lambda < 1$. Then:*

1. *$T$ has a unique fixed point $x^* \in X$.*
2. *For any initial point $x_0 \in X$, the sequence $\{x_n\}$ defined by $x_{n+1} = T(x_n)$ converges to $x^*$.*
3. *The error estimate is:*

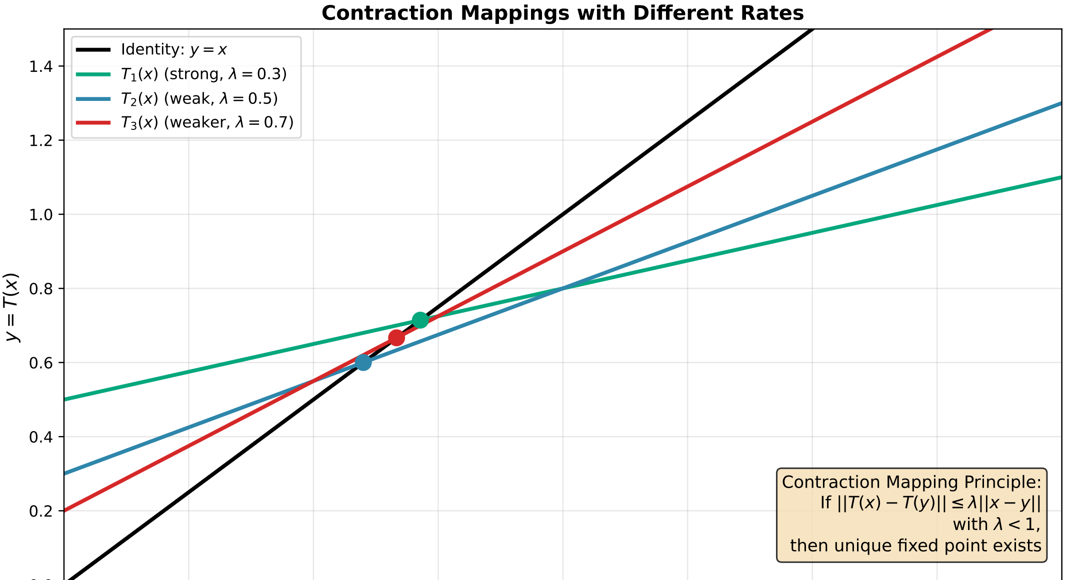
$$\|x_n - x^*\| \leq \lambda^n \|x_0 - x^*\|.$$

## Corollary (A Priori Error Bound)

*Before computing $x_n$, we can estimate:*

$$\|x_n - x^*\| \leq \frac{\lambda^n}{1 - \lambda} \|x_1 - x_0\|.$$

# Visual Illustration of Contraction Mappings



# Linear and Nonlinear Convergence I

**Definition (Order of Convergence)**

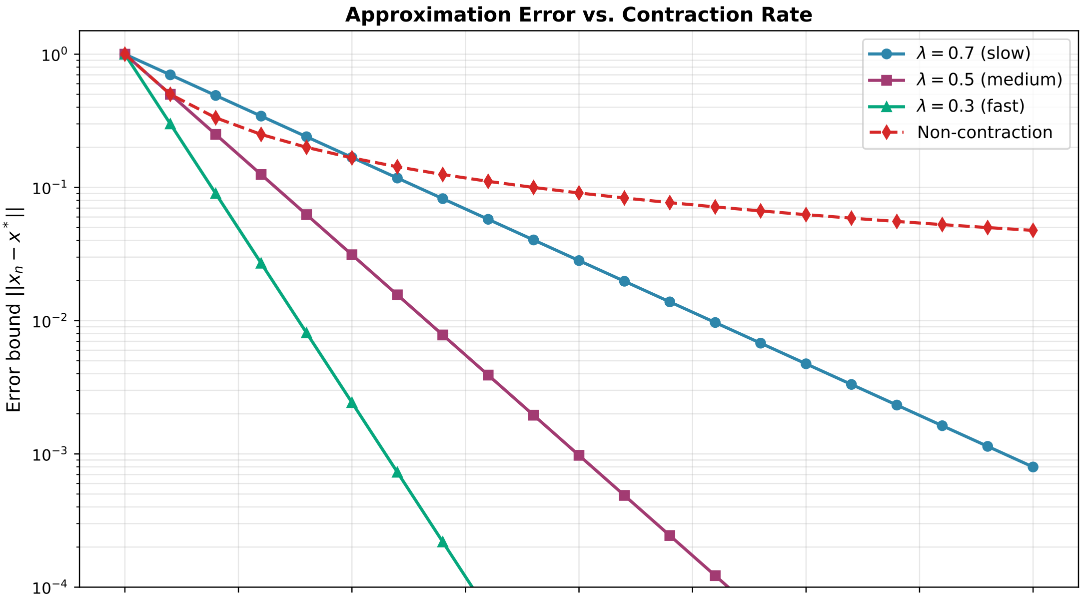
A sequence $\{x_n\}$ converges to $x^*$ with **order** $p$ if:

$$\|x_{n+1} - x^*\| \leq C\|x_n - x^*\|^p$$

for some constant $C > 0$ and all $n$ sufficiently large.

- **Linear convergence:** $p = 1$ with $0 < C < 1$
- **Superlinear convergence:** $p = 1$ with $C = 0$
- **Quadratic convergence:** $p = 2$
- **Superquadratic convergence:** $p > 2$

# Linear and Nonlinear Convergence II



# Error Bounds and Stopping Criteria

## Residual Error

At iteration $m$, define the residual error:

$$R_m = \|x_{m+1} - T(x_m)\| = \|x_{m+1} - x_{m+1}\| = 0 \quad \text{(by construction)}$$

More generally, we measure:

$$R_m = \|F(x_m)\|$$

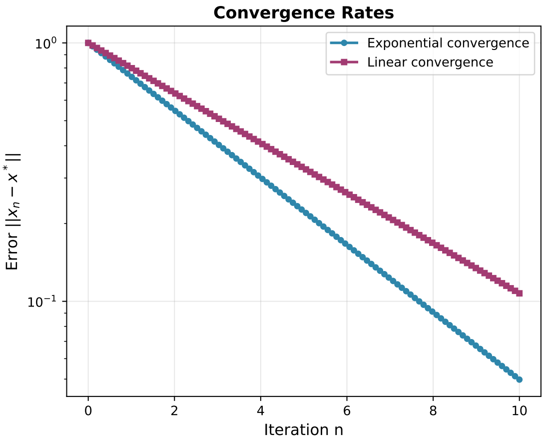
where $F(x) = x - T(x)$ is the **residual operator**.

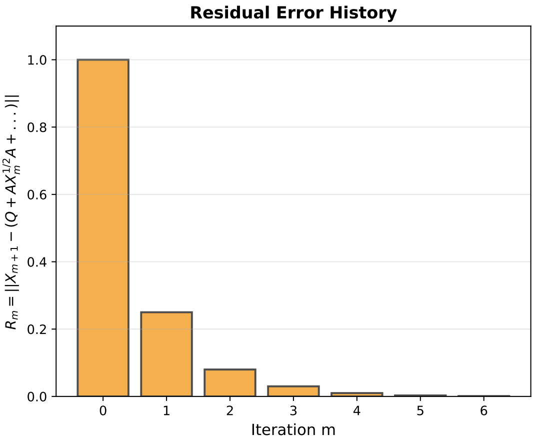
## Stopping Criteria

Common criteria to terminate iteration:

1. $\|x_{n+1} - x_n\| < \epsilon$ (successive approximation)
2. $\|F(x_n)\| < \epsilon$ (residual)
3. $n > N_{\max}$ (maximum iterations)
4. Relative error: $\frac{\|x_{n+1} - x_n\|}{\|x_{n+1}\|} < \epsilon_{\text{rel}}$

# Example: Convergence Behavior




The left panel shows exponential vs. linear convergence rates. The right panel displays residual error history over iterations.

# Mann Iteration

## Definition (Mann Iteration (1953))

For a mapping $T : D \to D$, the Mann iteration is defined by:

$$x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T(x_n), \quad n = 0, 1, 2, \ldots$$

where $\{\alpha_n\}$ is a sequence of parameters with $0 \leq \alpha_n \leq 1$.

## Properties

When $\alpha_n = 1$: recovers Picard iteration.

When $0 < \alpha_n < 1$: provides convex combination (damping).

Choice of $\{\alpha_n\}$ affects convergence:

- Constant $\alpha_n = \alpha$: easier analysis
- Variable $\{\alpha_n\}$: may accelerate convergence

## Theorem (Convergence of Mann Iteration)

# Mann Iteration Example

## Python Implementation

```
import numpy as np

def mann_iteration(T, x0, alpha=0.5, max_iter=100, tol=1e-8):
    """Mann iteration for fixed point approximation"""
    x = x0
    history = [x]

    for n in range(max_iter):
        T_x = T(x)
        x_new = (1 - alpha) * x + alpha * T_x

        if np.linalg.norm(x_new - x) < tol:
            print(f"Converged in {n+1} iterations")
            break

        x = x_new
        history.append(x)

    return x, np.array(history)

# Example: T(x) = 0.5*x + 0.3*sin(x)
T = lambda x: 0.5*x + 0.3*np.sin(x)
x_fixed, history = mann_iteration(T, x0=0.0)
```

# Ishikawa Iteration

## Definition (Ishikawa Iteration (1974))

The Ishikawa iteration is a two-step scheme:

$$y_n = (1 - \beta_n)x_n + \beta_n T(x_n), \tag{1}$$
$$x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T(y_n), \tag{2}$$

where $\{\alpha_n\}$ and $\{\beta_n\}$ are parameter sequences.

## Properties

Generalizes Mann iteration (recovers it when $\beta_n = 0$).

Potential for faster convergence with two parameters.

More computational cost (two applications of $T$ per iteration).

## Comparison of Methods

| Method | Iterations | Evaluations of $T$ |
|---|---|---|

# Noor Iteration

## Definition (Noor Iteration (1997))

A three-step iteration scheme:

$$y_n = (1-\beta_n)x_n + \beta_n T(x_n), \tag{3}$$

$$z_n = (1-\gamma_n)y_n + \gamma_n T(y_n), \tag{4}$$

$$x_{n+1} = (1-\alpha_n)z_n + \alpha_n T(z_n), \tag{5}$$

## Properties

Potentially faster convergence with three parameters.

Increased computational cost (3 evaluations of $T$ per iteration).

Useful for certain nonlinear operators and nonexpansive mappings.

# Two-Point Boundary Value Problems I

**Problem**

*Consider the boundary value problem:*

$$x''(t) = f(t, x(t), x'(t)), \quad 0 \leq t \leq T, \tag{6}$$

$$x(0) = \alpha, \quad x(T) = \beta. \tag{7}$$

# Two-Point Boundary Value Problems II

## Solution

*Transform to an integral equation:*

$$x(t) = \alpha + \frac{(\beta - \alpha)t}{T} - \int_0^T G(t,s) f(s, x(s), x'(s))\, ds,$$

*where* $G(t,s)$ *is the Green's function:*

$$G(t,s) = \begin{cases} \frac{s(T-t)}{T} & 0 \leq s \leq t \leq T \\ \frac{t(T-s)}{T} & 0 \leq t \leq s \leq T \end{cases}$$

# Two-Point Boundary Value Problems III

## Fixed Point Formulation

Define the operator $F : C^1[0, T] \to C^1[0, T]$ by:

$$(Fx)(t) = \alpha + \frac{(\beta - \alpha)t}{T} - \int_0^T G(t,s) f(s, x(s), x'(s))\, ds.$$

The boundary value problem solution is a fixed point of $F$:

$$F(x) = x \quad \Longleftrightarrow \quad x \text{ solves the BVP.}$$

## Theorem

*Under appropriate boundedness and continuity conditions on $f$ and the Green's function, $F$ is continuous and compact. By Schauder's fixed point theorem, $F$ has at least one fixed point.*

# Periodic Solution Problem

## Problem

*Find periodic solutions to:*

$$\frac{dx}{dt} = f(t, x(t)), \quad x(0) = x(p), \tag{8}$$

*where $p > 0$ is the period and $f(t, x)$ is periodic in $t$ with period $p$.*

## Solution

*Define the operator $F : C[0, p] \to C[0, p]$ by:*

$$(Fx)(t) = x_0 + \int_0^t f(s, x(s))\, ds,$$

*where the periodicity condition imposes additional constraints.*
*The Poincaré operator (period map) can be analyzed to show existence of periodic solutions via fixed point theory.*

# Hammerstein Integral Equations

## Definition (Hammerstein Equation)

An equation of the form:

$$x(t) = \lambda \int_a^b K(t,s) f(s, x(s))\, ds + g(t),$$

where:

- $K(t,s)$ is the kernel (typically continuous)
- $f(s,x)$ is a nonlinearity
- $\lambda$ is a parameter
- $g(t)$ is a given function

## Fixed Point Formulation

The Hammerstein equation can be written as a fixed point problem:

$$x = T(x),$$

# Hammerstein Equation: Numerical Example

## Example Setup

Consider:

$$x(t) = \lambda \int_0^1 (t + s) \exp(-x(s)) \, ds + t$$

## Python Solution

```
from scipy.integrate import quad

def hammerstein_operator(x_func, lam=0.5):
    """Hammerstein operator application"""
    def T(t):
        integrand = lambda s: (t + s) * np.exp(-x_func(s))
        result, _ = quad(integrand, 0, 1)
        return lam * result + t
    return T

# Fixed point iteration
def solve_hammerstein(lam, max_iter=50):
    # Start with x_0(t) = t
    x = lambda t: t
```

# Urysohn Integral Equations

## Definition (Urysohn Equation)

An equation of the form:

$$x(t) = \int_{\Omega} f(t, s, x(s))\, ds,$$

where $f : \Omega \times \Omega \times \mathbb{R} \to \mathbb{R}$ is a given kernel.

The Hammerstein equation is a special case where $f(t, s, x) = K(t, s)g(x)$.

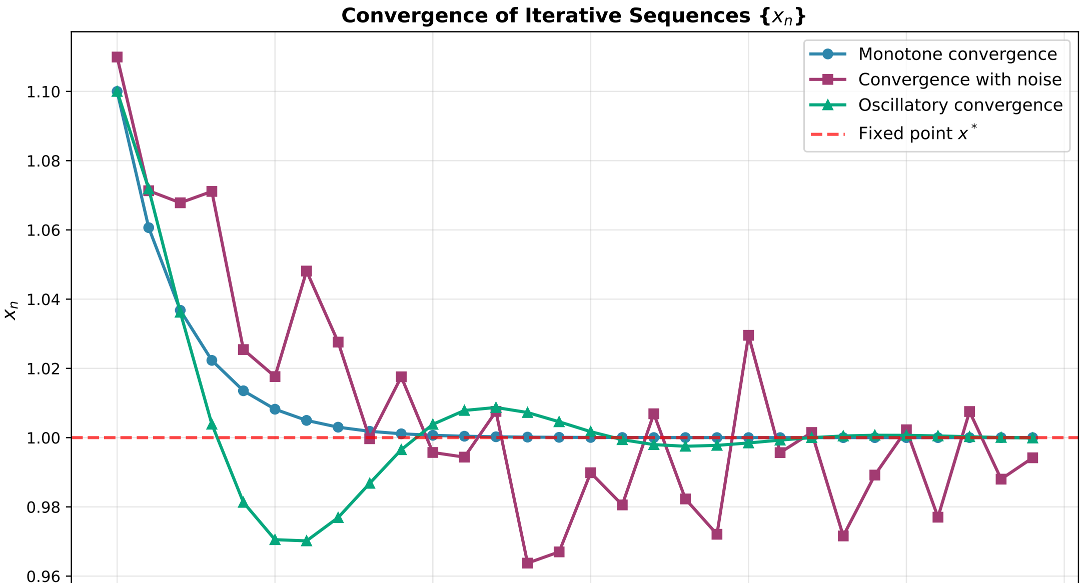
## Theorem (Existence of Solutions)

*If:*

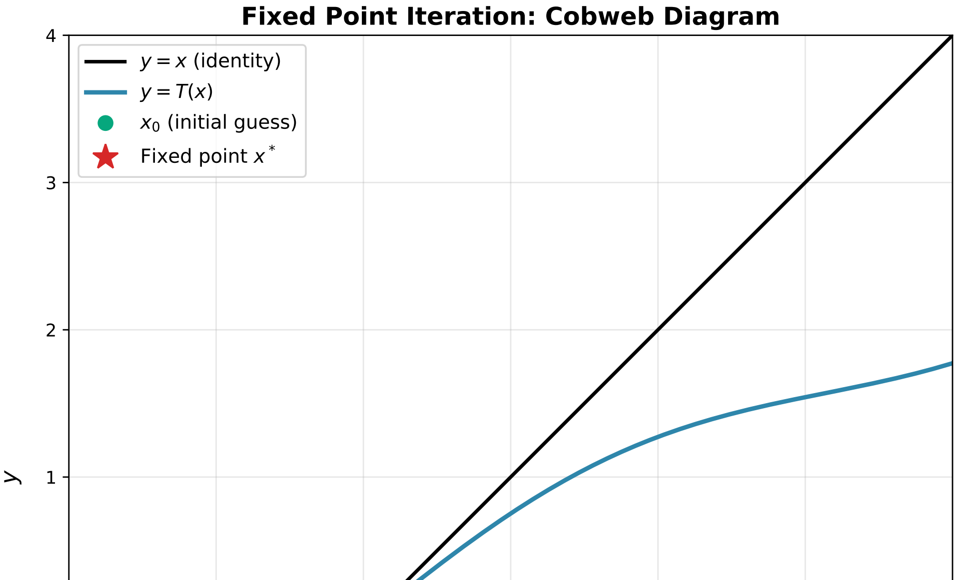
1. *$f$ is continuous on $\Omega \times \Omega \times \mathbb{R}$,*
2. *$f$ satisfies a Lipschitz condition in the third variable,*
3. *$\int_{\Omega} |f(t, s, 0)|\, ds \leq M$ for some constant $M$,*

*then the Urysohn equation has at least one solution in $C(\Omega)$.*

# Sequence Convergence Behaviors



# Fixed Point Iteration Visualization



# Computational Example: Matrix Fixed Point

## Problem

Solve $X = Q + AX^{1/2}A^* + BX^{-1}B^*$ using Picard iteration.

```
import numpy as np
from scipy.linalg import sqrtm

def matrix_fixed_point(Q, A, B, X0, max_iter=100, tol=1e-8):
    """Solve matrix fixed point equation iteratively"""
    X = X0.copy()

    for m in range(max_iter):
        X_sqrt = sqrtm(X)
        X_inv = np.linalg.inv(X)

        # Picard step: X_{m+1} = T(X_m)
        X_new = Q + A @ X_sqrt @ A.T + B @ X_inv @ B.T

        # Compute residual
        residual = np.linalg.norm(X_new - X)

        if residual < tol:
            print(f"Converged in {m+1} iterations")
            return X_new

```

# Nonexpansive Mappings

## Definition (Nonexpansive Mapping)

A mapping $T : D \to X$ is **nonexpansive** if:

$$\|T(x) - T(y)\| \leq \|x - y\| \quad \text{for all } x, y \in D.$$

**Key difference from contractions:**

- Nonexpansive: contraction rate $\lambda = 1$ (not strictly less than 1).
- Banach theorem does not directly apply!
- Requires additional structure (e.g., compactness, normal structure).

## Theorem (Schauder Fixed Point Theorem)

*If $T : K \to K$ is continuous and $K$ is a nonempty compact convex subset of a Banach space, then $T$ has a fixed point in $K$.*

# Quasi-Nonexpansive Mappings

## Definition (Quasi-Nonexpansive)

A mapping $T$ is quasi-nonexpansive if it has a fixed point $x^*$ and:

$$\|T(x) - x^*\| \leq \|x - x^*\| \quad \text{for all } x \in D.$$

## Properties

Weaker than nonexpansive (doesn't require $\|T(x) - T(y)\| \leq \|x - y\|$ for all $x, y$).

Guarantees monotone convergence to the fixed point.

Examples: projections onto closed convex sets.

## Theorem

*If $T$ is quasi-nonexpansive and Mann iteration is used with appropriate parameters $\{\alpha_n\}$, then $\{x_n\}$ converges to the fixed point.*

# Hybrid Methods and Operator Splitting

## Motivation

For complex operators $T = T_1 + T_2$ or $T = T_1 \circ T_2$, apply iterations to individual components.

## Definition (Operator Splitting)

Instead of:

$$x_{n+1} = T(x_n),$$

use:

$$x_{n+1} = T_1(T_2(x_n)), \quad \text{(Alternating)} \tag{9}$$

$$x_{n+1} = \lambda T_1(x_n) + (1-\lambda) T_2(x_n), \quad \text{(Convex combination)} \tag{10}$$

## Applications

Useful for:

- Proximal operator splitting in optimization.

# Summary of Key Results

1. **Banach Contraction Principle**: $\lambda$-contractions have unique fixed points reachable by Picard iteration with exponential convergence.
2. **Iterative Methods**: Mann, Ishikawa, and Noor iterations generalize Picard iteration with damping parameters for improved convergence.
3. **Differential Equations**: Boundary value and periodic solution problems reduce to fixed point equations via Green's functions.
4. **Integral Equations**: Hammerstein and Urysohn equations are fixed point problems in function spaces.
5. **Nonexpansive and Quasi-Nonexpansive**: Extended theory for mappings not satisfying strict contractiveness.

# Applications Across Disciplines

## Numerical Analysis

- Solving nonlinear equations (Newton's method, fixed point iteration)
- Eigenvalue problems and spectral computations
- Matrix equations and tensor computations

## Optimization

- Proximal methods and forward-backward splitting
- Alternating direction method of multipliers (ADMM)
- Stochastic gradient descent and online learning

## Engineering and Physics

- Control system design and feedback loops
- Chemical reactor models
- Fluid dynamics and partial differential equations

# Open Problems and Future Directions

1. **Acceleration Techniques**: How to enhance convergence rates using preprocessing, extrapolation, or Krylov subspace methods?
2. **Nonlinear Operators**: Theory for more general nonlinear structures beyond contractions and nonexpansive mappings.
3. **Stochastic Algorithms**: Convergence analysis for randomized and stochastic fixed point iterations.
4. **Distributed Computing**: Fixed point methods for parallel and distributed computing in large-scale problems.
5. **Machine Learning**: Applications to deep learning training algorithms and neural network optimization.

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